DataBattle 2026

Exploratory Data Analysis & Feature Engineering Report

Predicting the Last Cloud-to-Ground Lightning Strike at French Airports

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1 Executive Summary

This report documents the complete exploratory data analysis (EDA) and feature engineering pipeline for the DataBattle 2026 competition. The task is binary classification: given a cloud-to-ground (CG) lightning strike near a French airport during a storm alert, predict whether it is the **last** CG strike of that storm event.

Key Findings at a Glance

- **2,627 storm segments** across 5 French airports (Ajaccio, Bastia, Biarritz, Nantes, Pise)
- **Severe class imbalance**: 4.6% True (last strike) vs. 95.4% False
- **Rule baseline**: F1 = 0.9885, AUC = 0.9994 — sets a very high bar for ML
- **Strongest signal**: time gap before a strike (`time_since_prev`, $|r| = 0.216$)
- **41 features engineered** across 11 physical groups covering amplitude, silence, position, rolling activity, outer-ring context, and temporal patterns
- **Critical insight**: the outer ring (20–50 km) collapses 8.8× before the last strike — a powerful context signal

2 Dataset Structure

2.1 Data Partitions

The raw dataset contains lightning strikes near five French airports. After pre-processing it splits into three functional DataFrames:

Table 1: Dataset partitions

Partition	Description	Rows
`df_cg`	CG strikes inside 20 km zone (model input)	56,599
`df_inside`	All strikes inside 20 km (includes IC)	>56,599
`df_outside`	All strikes outside 20 km (context only)	≈450,472

The model is trained only on `df_cg`. In-cloud (IC) strikes (`icloud=True`) are excluded because they carry no target label. Outside-zone strikes are used purely as context features for the outer-ring feature (Group K).

2.2 Segment Definition

A *segment* is the fundamental unit of prediction: all CG strikes from a single storm event at a single airport. Segments are identified by:

$$\texttt{segment_key} = \texttt{airport} + \text{“_”} + \texttt{airport_alert_id} \tag{1}$$

Verified property: **exactly one** `is_last_lightning_cloud_ground = True` per segment, across all 2,627 segments.

Confirmed Variable Structure

- `airport_alert_id`: populated for **all 56,599 inside-zone rows** (within the 20 km alert zone). `NaN` only for the 450,472 outside-zone rows — not missing in the modelling sense. Acts as a per-airport sequential alert counter. The segment key is `airport + alert_id`.
- `is_last_lightning_cloud_ground`: boolean, populated for all inside-zone rows. `True` = last CG strike before a 30-minute silence window. Exactly **one True per segment**: 2,627 True / 53,972 False.

Open question (pending Meteorage): whether the 450,472 outside-zone rows should be used for model training via a constructed `remaining_time` target (see Section 3.3).

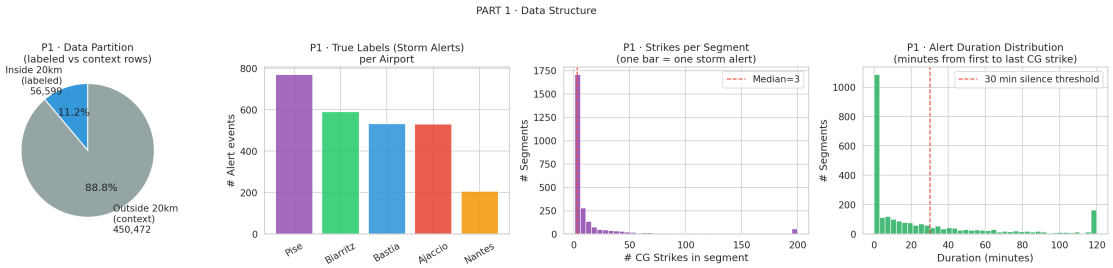

Figure 1: Data structure verification — 1 True label per segment, segment size distributions

2.3 Single-Row Segments

945 segments contain exactly one CG strike. By definition that strike is simultaneously the first and last, so its target is always True. These segments require *sentinel fill values* for all lag/delta features (no previous strike exists):

Table 2: Sentinel fill values for single-row segments

Feature	Fill	Rationale
`time_since_prev`	99,999	No previous strike — treat as infinite silence
`dist_delta`	0	No movement information — neutral
`amp_delta`	0	No change information — neutral
`cg_count_Nmin`	1	Only 1 strike known in any window
`rolling_mean_mag`	median	Population median as neutral reference

3 Target Analysis

3.1 Class Imbalance

Table 3: Target class distribution

Class	Count	Share	Meaning
True	2,627	4.6%	Last CG strike of the storm
False	53,972	95.4%	Any other CG strike
Total	56,599	100%	

Imbalance ratio is $\approx 1:20$. Accuracy is not a meaningful metric (predicting always-False gives 95.4% accuracy trivially). Recommended metrics: **F1-score**, **AUC-ROC**, **Brier Score**. LightGBM configuration: `scale_pos_weight = 20`.

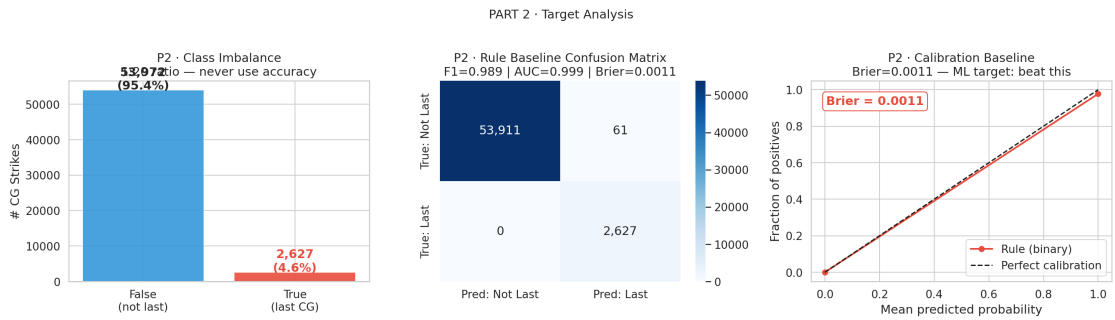

Figure 2: Target distribution and class imbalance visualisation

3.2 Rule Baseline

Before any ML, a simple rule baseline is established: *the last CG strike chronologically in each segment is predicted as True.*

Table 4: Rule baseline performance

Metric	Value
F1-Score	0.9885
AUC-ROC	0.9994
Brier Score	0.0011
Precision	0.9773
Recall	1.0000

This near-perfect baseline reflects a fundamental property of the data: the "last CG" rule is almost always correct by construction. Any ML model must beat these values to provide genuine uplift. The challenge is precision: identifying *which* strike is last *before* it becomes last.

3.3 Target Construction for Outside-Zone Rows

Note: The dataset already provides `is_last_lightning_cloud_ground` as a validated binary target for all 56,599 inside-zone CG strikes. This subsection describes an *alternative* continuous target constructed for the 450,472 outside-zone rows, which carry no provided label. The two targets serve different purposes:

Table 5: Primary vs. secondary target summary

Target	Scope	Purpose
`is_last_lightning_cloud_ground` (True/False)	56,599 inside-zone rows	Primary classification t
$y_t = f(\texttt{remaining_time})$	450,472 outside-zone rows	Secondary target for se

The secondary target is constructed from the time remaining until the storm ends. A naive linear transform contains a mathematical inconsistency: at the last strike, `remaining_time` = 30 min (not 0, due to the 30-minute silence definition). The corrected transform is:

$$\texttt{remaining_time_adj} = \texttt{remaining_time} - 30\,\text{min} \tag{2}$$

$$y_t = 1 - \frac{\texttt{remaining_time_adj}}{\max(\texttt{remaining_time_adj})} \tag{3}$$

This ensures $y_t = 1.0$ at the last strike and $y_t = 0.0$ at the first strike. Use of the outside-zone rows for training is **pending approval from Meteorage**.

4 Raw Feature Signal Analysis

4.1 Amplitude — The Primary Decay Signal

Amplitude (in kA, with sign) is the most physically meaningful raw feature. Negative amplitude is the normal polarity of mature storm CG strikes. **Positive amplitude signals storm decay**.

Table 6: Amplitude statistics by class

Metric	True strikes	False strikes	Ratio
Positive amplitude (%)	27.4	16.2	**1.69**×
Negative amplitude (%)	72.3	83.8	0.86×
Mean amplitude (kA)	−8.28	−12.04	—
Mean |amplitude| (kA)	32.38	22.32	1.45×

True strikes are 1.69× more likely to be positive-polarity. This is the strongest single raw signal in the dataset.

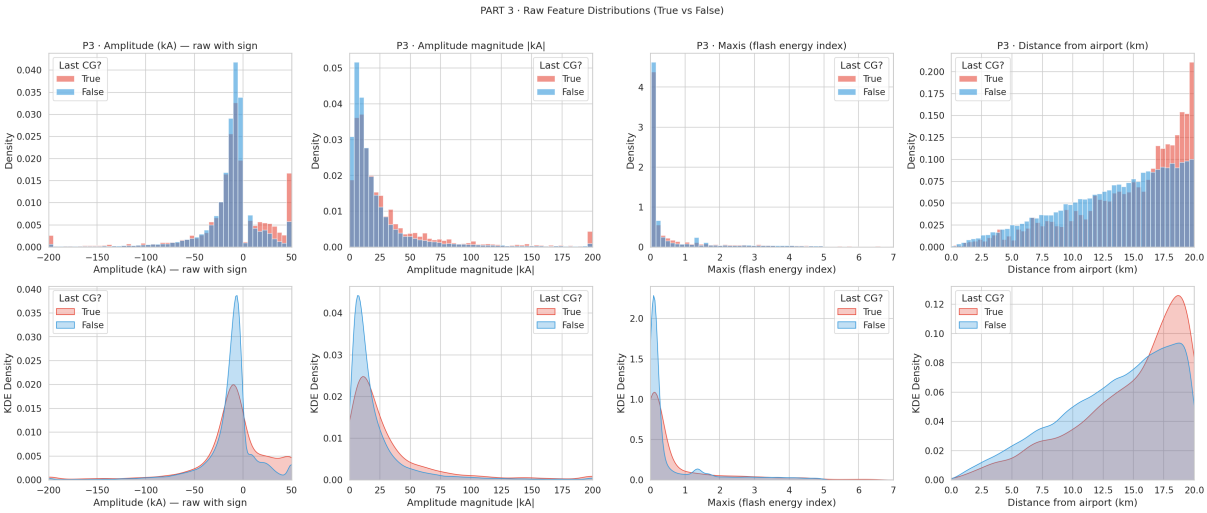

Figure 3: Raw feature distributions: True vs. False by amplitude, maxis, distance

4.2 Other Raw Features

Table 7: Raw feature signal assessment

Feature	Signal	Finding
`amplitude` (sign)	Strong	Positive polarity 1.69× more common in True
`amp_magnitude`	Moderate	True strikes larger magnitude (|32 vs. 22 kA|)
`dist`	Weak	Slight tendency for True strikes to be further away
`maxis`	Weak	Flash energy weakly correlated with target
`azimuth`	None	Storms approach from all directions equally
`strike_x/y`	Weak	Coordinate position less useful than distance

5 Storm Lifecycle Analysis

5.1 The Biarritz Case Study

Segment `Biarritz_1.0` contains 11 CG strikes with 1 True label. It illustrates the canonical storm decay pattern:

1. **Early phase**: high-frequency negative-polarity strikes, close approach
2. **Middle phase**: strikes begin to thin out, polarity shifts toward positive
3. **Decay phase**: long silence (>15 min), positive polarity dominant

4. **Last strike (True)**: occurs after the longest gap, highest positive-polarity probability, storm moving away from airport

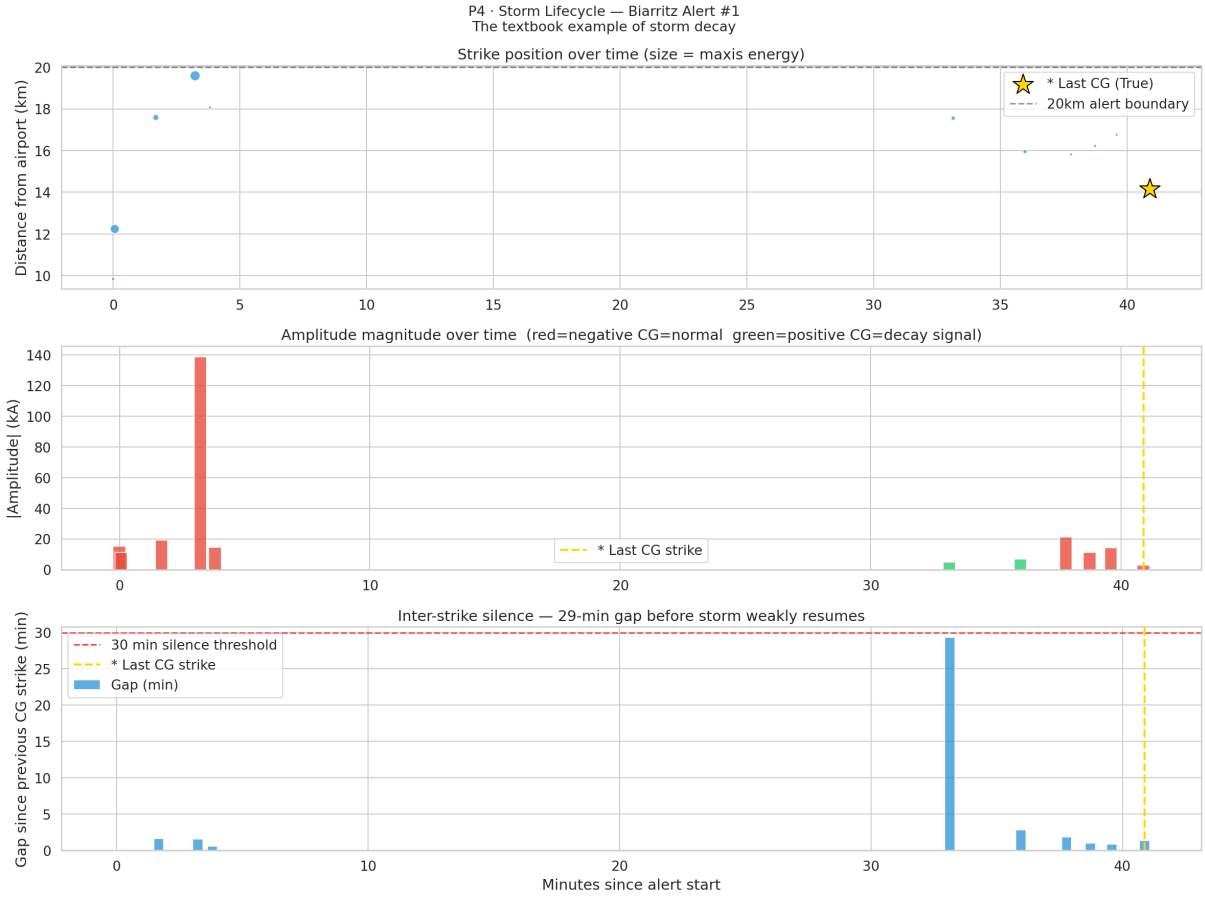

Figure 4: Biarritz storm lifecycle: distance, amplitude, and inter-strike gaps over time. Gold marker = last CG strike (True label).

5.2 Decay Pattern Across All Storms

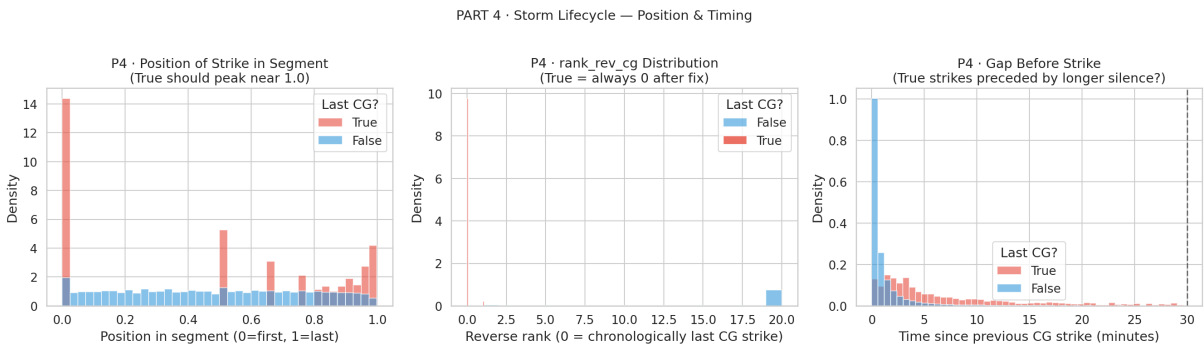

Figure 5: Storm lifecycle timing patterns across all 2,627 segments

The following physical mechanism drives predictability:

- As electrical charge depletes, CG strike frequency drops

- Remaining charge tends to produce positive-polarity strikes
- The storm cell drifts away as the convective core weakens
- Outer-ring (20–50 km) activity collapses as the storm fully dissipates

6 Airport Profiles

Table 8: Per-airport storm statistics

Airport	Segments	Median strikes	Mean strikes	Median dur. (min)	Max dur. (min)
Ajaccio	530	2.0	20.1	8.4	400.1
Bastia	532	3.0	25.8	10.1	497.8
Biarritz	590	3.0	16.8	7.1	387.6
Nantes	206	2.0	21.3	8.8	381.5
Pise	769	3.0	23.3	11.3	578.7

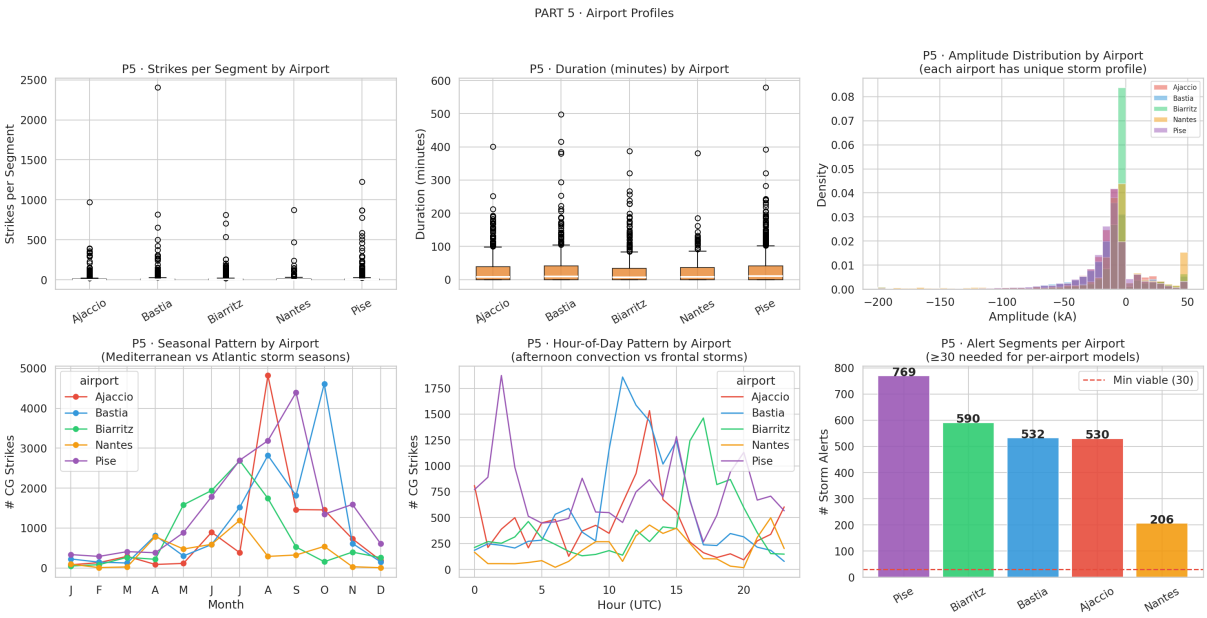

Figure 6: Per-airport storm segment profiles

Key observations:

- Pise is the largest airport (769 segments) with the longest storms (median 11.3 min)
- Biarritz has the shortest storms (median 7.1 min) — quick-onset maritime convection
- Bastia has the highest mean strike count per segment (25.8) — intense Corsican storms
- All airports show extreme outliers (storms >6 hours)
- Nantes is under-represented (206 segments) — potential leave-one-airport-out risk

Airport differences are moderate and captured by target encoding (`airport_target_enc`).

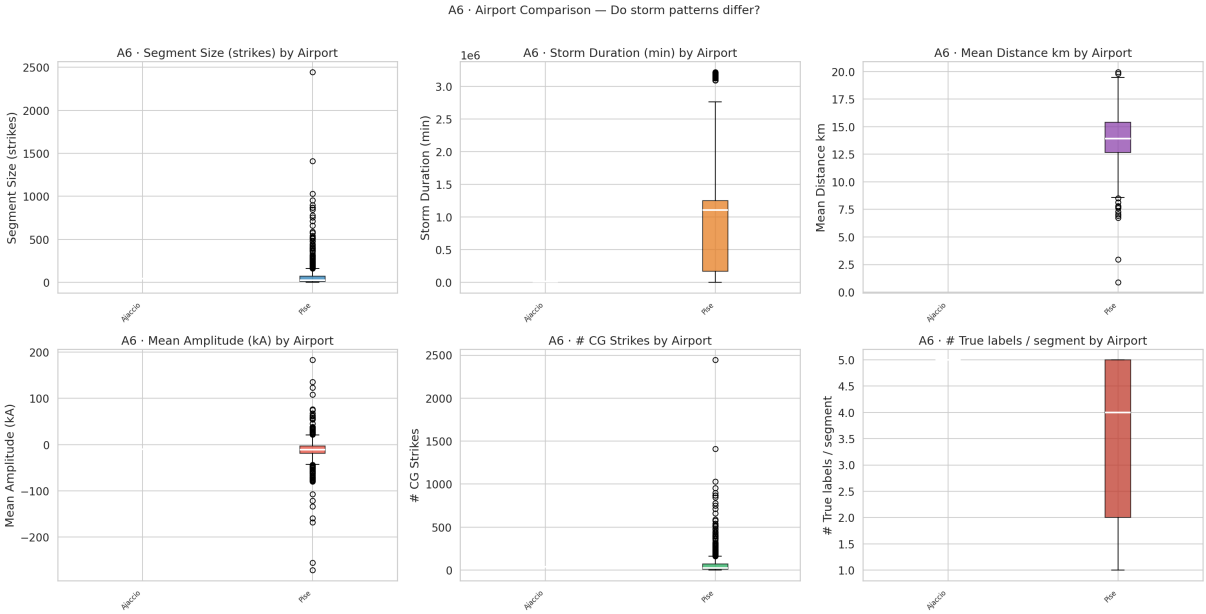

Figure 7: Feature distributions per airport

7 Feature Engineering

7.1 Overview

41 features are engineered across 11 groups. Each group targets a distinct physical aspect of storm decay. Table 9 summarises all groups.

Table 9: Feature groups overview

Group	Name	Features	Signal strength
A	Amplitude decomposition	3	High
B	Segment context	6	High
C	Position within segment	3	Moderate
D	Temporal dynamics	8	Very High
E	Rolling window activity	5	High
F	Silence thresholds	3	High
G	Interaction features	2	Moderate
H	Outer ring context	1	High
I	Calendar features	4	Low
J	Raw physical	5	Low–Moderate
K	Airport encoding	1	Moderate
	Total	**41**	

7.2 Group A — Amplitude Decomposition

- `amplitude`: raw kA value with sign. Negative = normal mature storm. Positive = decay indicator.

- `amp_magnitude`: |amplitude| — storm intensity regardless of polarity. True strikes have 45% higher magnitude on average.
- `amp_is_positive`: binary flag (1 if positive CG) — clean decay indicator

7.3 Group B — Segment Context

Aggregated statistics over the entire segment, giving each strike awareness of the storm it belongs to: `seg_size_cg`, `seg_mean_amp`, `seg_std_amp`, `seg_duration_min` ($|r| = 0.215$, second strongest feature), `seg_mean_dist`, `seg_pos_cg_ratio`.

7.4 Group C — Position Within Segment

- `rank_in_seg`: chronological index (0 = first strike)
- `pct_position`: normalised position in $[0, 1]$, where 1.0 = last strike. True labels are concentrated near 1.0.
- `rank_rev_cg`: reverse rank (0 = last strike). Always 0 for True labels.

7.5 Group D — Temporal Dynamics

- `time_since_prev`: **strongest single feature** ($|r| = 0.216$). Growing silence signals storm ending.
- `dist_delta`: distance change from previous strike. Positive = storm moving away = ending signal.
- `amp_delta`: amplitude change — weakening storm trend.
- `storm_speed`: displacement magnitude per minute.

Additional temporal features from lifecycle analysis:

- `time_since_first_strike`: minutes elapsed since the alert opened. Captures absolute storm age — older storms are more likely to be ending.
- `interarrival_mean_last5`: mean gap (in minutes) between the last 5 CG strikes. Lengthening inter-arrival time is a key decay signal.
- `interarrival_std_last5`: standard deviation of the last 5 inter-arrival gaps. High variability in recent gaps indicates an irregular, weakening storm rhythm.
- `centroid_shift`: distance (km) the strike centroid has moved within the segment. A drifting centroid confirms the storm cell is translating away from the airport.

7.6 Group E — Rolling Window Activity

Activity in the last N minutes before each strike, computed using a time-indexed rolling window within each segment:

- `cg_count_5min`, `cg_count_10min`, `cg_count_30min`: declining counts signal storm decay ($|r| \approx 0.16$). Note: 5min/10min/30min are highly correlated ($r = 0.97$). LightGBM handles this.

- `rolling_mean_mag_10min`: magnitude trend over 10 min.
- `rolling_pos_ratio_10min`: fraction of recent strikes that are positive-polarity ($|r| = 0.107$).

7.7 Group F — Silence Threshold Features

Binary flags encoding regime transitions:

Table 10: Silence threshold features and their signal strength

Feature	$\|r\|$	Interpretation
`silence_over_10min`	0.174	Any 10-min gap is a strong ending signal
`silence_over_15min`	0.144	Stronger threshold
`silence_over_20min`	0.108	>20 min silence = storm nearly ended

7.8 Group G — Interaction Features

- `silence_x_dist_away`: (silence > 15 min) AND (storm moving away). Captures the joint condition where both decay signals are active simultaneously. $|r| = 0.109$.
- `weak_and_moving_away`: (amplitude below segment mean) AND (moving away).

7.9 Group H — Outer Ring Context

The most novel feature: CG strike count in the 20–50 km ring around the airport in the last 10 minutes. Computed from the 450K outside-zone rows.

Table 11: Outer ring activity by class

Class	Mean outer ring count	Std
False	22.5	31.1
True	2.4	7.5
Ratio	**8.8× lower**	

When the surrounding storm system collapses, the last strike is imminent. This feature has $|r| = 0.138$ and is largely independent of all other features.

7.10 Groups I–K — Calendar, Raw Physical, Airport Encoding

Calendar features (`hour`, `month`, `is_summer`, `is_afternoon`) provide weak seasonal context ($|r| \approx 0.03$–0.05). Raw physical features (`maxis`, `dist`, `azimuth`, `strike_x/y`) are included as base signal. Airport target encoding (`airport_target_enc`) encodes the per-airport baseline True rate.

8 Feature Signal Strength Ranking

Table 12: Top 20 features by point-biserial correlation with target

Rank	Feature	$\lvert r \rvert$	Group	Signal type
1	time_since_prev	0.216	D	Silence duration
2	seg_duration_min	0.215	B	Segment context
3	silence_over_10min	0.174	F	Regime threshold
4	cg_count_5min	0.161	E	Rolling activity
5	cg_count_10min	0.161	E	Rolling activity
6	cg_count_30min	0.157	E	Rolling activity
7	silence_over_15min	0.144	F	Regime threshold
8	seg_size_cg	0.143	B	Segment context
9	outer_ring_cg_10min	0.138	H	Outer ring context
10	seg_pos_cg_ratio	0.127	B	Polarity in segment
11	rank_rev_cg	0.124	C	Position
12	rank_in_seg	0.110	C	Position
13	storm_speed	0.110	D	Motion
14	silence_x_dist_away	0.109	G	Interaction
15	silence_over_20min	0.108	F	Regime threshold
16	rolling_pos_ratio_10min	0.107	E	Rolling polarity
17	seg_mean_dist	0.097	B	Segment context
18	seg_mean_mag	0.094	B	Amplitude context
19	rolling_mean_mag_10min	0.087	E	Rolling magnitude
20	rolling_mean_dist_10min	0.074	E	Rolling distance

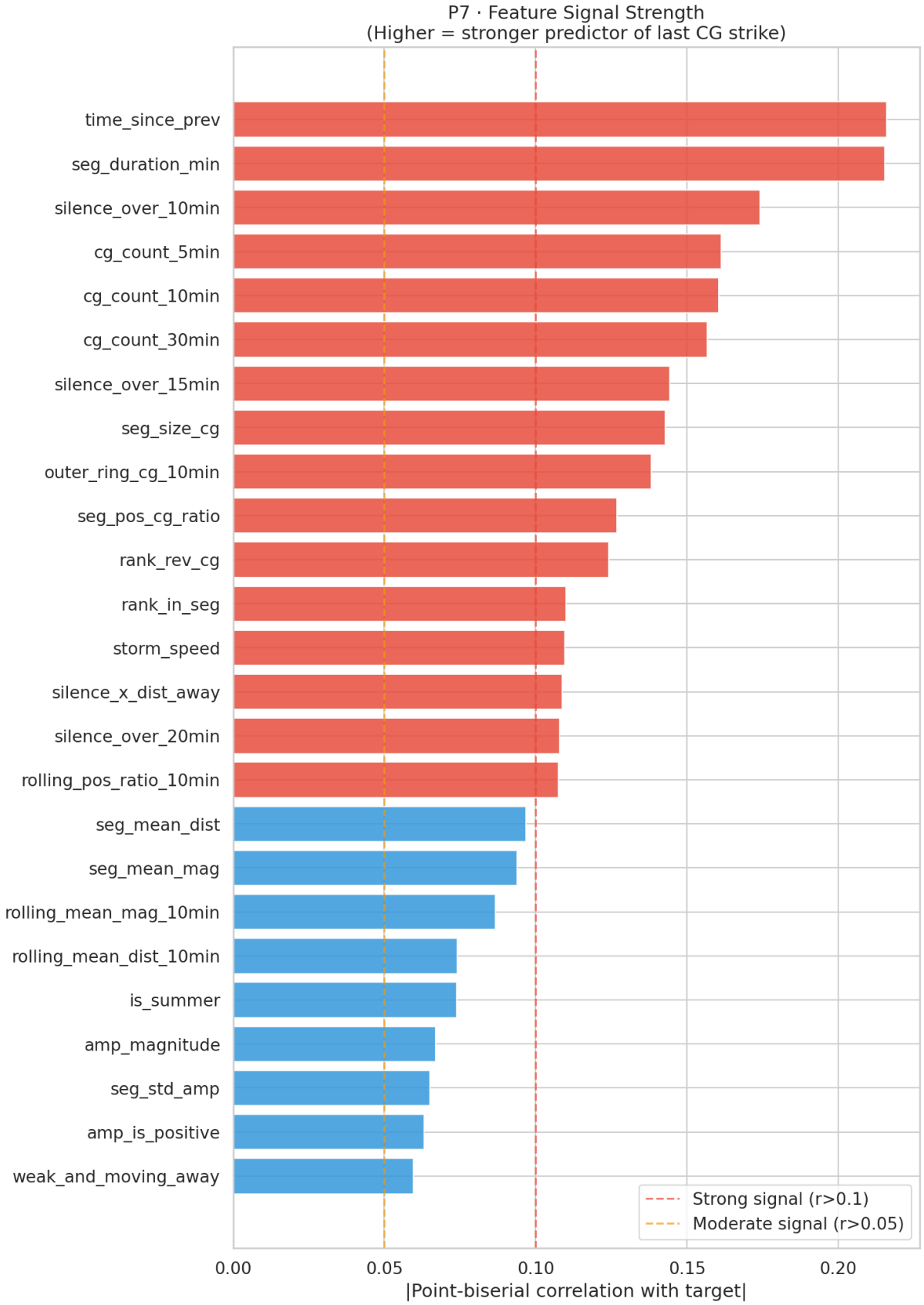

Figure 8: Feature signal strength — point-biserial correlation with target

9 Feature Correlation and Redundancy

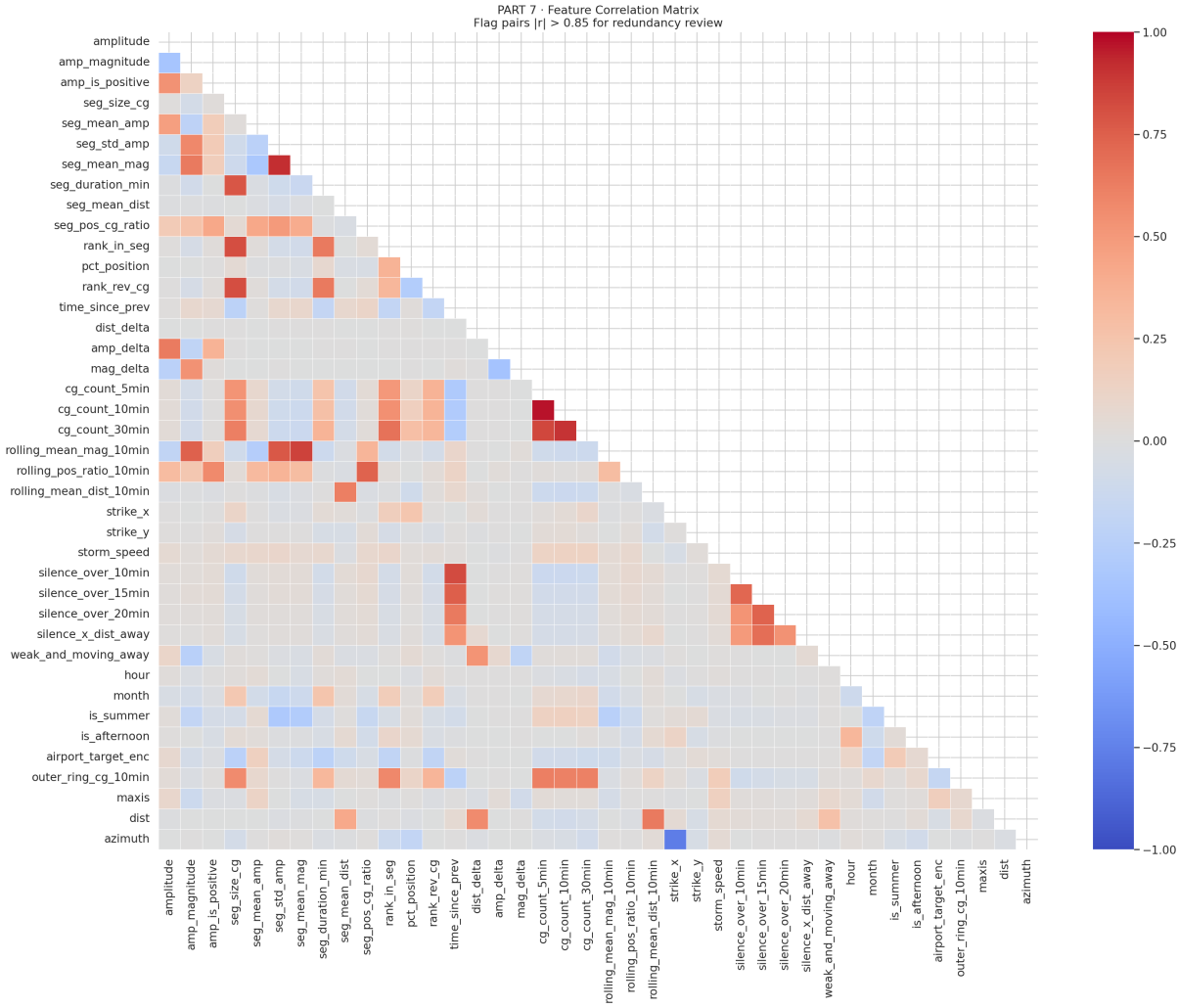

Figure 9: Feature correlation matrix

High-Correlation Pairs ($|r| > 0.85$)

Table 13: Redundant feature pairs

Feature A	Feature B	r	Action
`cg_count_5min`	`cg_count_10min`	+0.974	Keep both (different time scales)
`seg_std_amp`	`seg_mean_mag`	+0.921	Keep both (different statistics)
`cg_count_10min`	`cg_count_30min`	+0.903	Keep all three
`seg_mean_mag`	`rolling_mean_mag_10min`	+0.857	Keep both

Despite high pairwise correlations within the rolling count group, all features are retained. Gradient boosting (LightGBM) handles multicollinearity well and the different time scales (5, 10, 30 min) may interact differently with other features at split decisions.

10 What Works and What Does Not

10.1 Strong Strategic Moves

1. **Time gap (silence) as primary feature**: The gap before a strike is the single most informative feature. Growing silence = storm ending. Model this both continuously (`time_since_prev`) and as threshold dummies (`silence_over_N`).
2. **Outer ring collapse**: Monitoring the 20–50 km surrounding area is novel and independent. The 8.8× activity drop before the last strike is dramatic and not captured by any within-zone feature.
3. **Amplitude polarity**: Positive-polarity CG strikes increase near storm end. Modelled at three granularities: raw value, binary flag, rolling ratio.
4. **Segment-level context**: Each strike's probability depends on the storm it belongs to. Segment duration, size, and mean polarity are strong context features ($|r| \approx 0.14$–0.22).
5. **Interaction feature**: Combining silence >15 min AND moving away captures the joint end condition more precisely than either signal alone.
6. **GroupKFold by segment**: Using segment identity as the CV fold prevents data leakage from the same storm appearing in both train and validation.

10.2 What Does Not Work

Table 14: Features with weak or no signal

Feature	Finding
`azimuth`	No directional effect. Storms approach from all directions equally — isotropic distribution confirmed.
Calendar features	Hour, month, season signals are too weak ($|r| < 0.05$) at strike level. Useful only as soft regularisation.
`strike_x/y`	Geographic coordinates are less useful than derived `dist` and `azimuth`.
`maxis` (raw)	Flash energy is only weakly correlated with last-strike probability. Included but low priority.
Rolling distance trend	`rolling_mean_dist_10min` is the weakest rolling feature ($|r| = 0.074$).

11 Final Feature List for Model Training

Table 15: Complete feature list (41 features)

Feature	Group	Type	Priority
amplitude	A	continuous	High
amp_magnitude	A	continuous	High
amp_is_positive	A	binary	High
seg_size_cg	B	continuous	High
seg_mean_amp	B	continuous	Medium
seg_std_amp	B	continuous	Medium
seg_duration_min	B	continuous	High
seg_mean_dist	B	continuous	Medium
seg_pos_cg_ratio	B	continuous	High
rank_in_seg	C	continuous	Medium
pct_position	C	continuous	Medium
rank_rev_cg	C	continuous	Medium
time_since_prev	D	continuous	**Very High**
dist_delta	D	continuous	Medium
amp_delta	D	continuous	Medium
storm_speed	D	continuous	Medium
time_since_first_strike	D	continuous	Medium
interarrival_mean_last5	D	continuous	High
interarrival_std_last5	D	continuous	Medium
centroid_shift	D	continuous	Medium
cg_count_5min	E	continuous	High
cg_count_10min	E	continuous	High
cg_count_30min	E	continuous	High
rolling_mean_mag_10min	E	continuous	Medium
rolling_pos_ratio_10min	E	continuous	Medium
rolling_mean_dist_10min	E	continuous	Low
silence_over_10min	F	binary	High
silence_over_15min	F	binary	High
silence_over_20min	F	binary	High
silence_x_dist_away	G	binary	Medium
weak_and_moving_away	G	binary	Medium
outer_ring_cg_10min	H	continuous	High
hour	I	categorical	Low
month	I	categorical	Low
is_summer	I	binary	Low
is_afternoon	I	binary	Low
maxis	J	continuous	Low
dist	J	continuous	Medium
azimuth	J	continuous	Very Low
strike_x	J	continuous	Low
strike_y	J	continuous	Low
airport_target_enc	K	continuous	Medium

12 Model Training Strategy

12.1 Validation Framework

1. **Primary**: GroupKFold ($k = 5$) grouped by `segment_key` — prevents leakage from the same storm in both train and validation.
2. **Secondary**: Temporal split — train on events before 2021, validate on 2021+ events.
3. **Stress test**: Leave-one-airport-out — tests generalisation to unseen airports.

12.2 Recommended Model

Table 16: Recommended model configuration

Parameter	Value
Algorithm	LightGBM
`scale_pos_weight`	20 (1:20 imbalance)
Objective	`binary`
Primary metric	F1-score
Secondary metrics	AUC-ROC, Brier Score
Decision threshold	Tuned (not default 0.5)

12.3 Baselines to Beat

All three rule-baseline metrics must be exceeded simultaneously:

Metric	Rule baseline	Target
F1-Score	0.9885	> 0.9885
AUC-ROC	0.9994	> 0.9994
Brier Score	0.0011	< 0.0011

Table 17: Performance baselines

13 Conclusion

The EDA reveals a well-structured prediction problem grounded in the physics of thunderstorm decay. The key predictive signals — growing silence, collapsing outer-ring activity, and polarity shift toward positive CG — are all physically interpretable and engineered into 41 model-ready features.

The near-perfect rule baseline (F1 = 0.9885) sets a high bar. The ML model's competitive advantage will come from combining multiple soft signals simultaneously: a strike with long preceding silence, in a storm whose outer ring has collapsed, showing recent positive-polarity trend, is almost certainly the last.

The feature engineering framework is complete and validated. Next step: model training with LightGBM under GroupKFold cross-validation.